UNIPOTENT SHEAR ON FIXED-INDEX SUBLATTICES: COMPLETE CYCLE DATA AND VALUATION STAIRCASES

ANONYMOUS

ABSTRACT. For a positive integer N, let $\mathcal{L}_N$ be the finite set of index-N sublattices of $\mathbb{Z}^2$. We determine the complete finite dynamics on $\mathcal{L}_N$ induced by the unipotent shear $U = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Unique column Hermite coordinates turn the action into a disjoint union of translations $b \mapsto b + N/a$ on $\mathbb{Z}/a\mathbb{Z}$, one for every divisor a of N. This gives every cycle, every fixed-point count, and the finite Artin–Mazur zeta function in closed form. For $N = p^r$, the fixed counts form a parity-sensitive valuation staircase and the zeta function has an explicit sparse prime-power product. Finally, the maximal cycle is unique and has length N, so the full temporal data recover the index. Exact integer controls enumerate every canonical phase and trace its orbit for $1 \le N \le 120$, with separate prime-power tests.

1. Introduction

Fix $N \ge 1$ and write

$$\mathcal{L}_N = \{L \le \mathbb{Z}^2 : [\mathbb{Z}^2 : L] = N\}.$$

The integral matrix

$$U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

lies in $\mathrm{SL}_2(\mathbb{Z})$ and therefore acts by the permutation $T_N(L) = UL$ of $\mathcal{L}_N$. Although U has infinite order on $\mathbb{Z}^2$, every T_N is a finite permutation. The purpose of this note is to give its complete temporal census.

The calculation has two ingredients. First, column Hermite normal form provides a canonical divisor-layer coordinate system; see, for example, Cohen [2, Chapter 2]. Second, on each layer the shear is a cyclic translation. The resulting formulas simultaneously describe cycle lengths, fixed points, and the Artin–Mazur zeta function [1]. The prime-power specialization exposes a sharp valuation staircase that is less visible in the undigested divisor sum.

Classical ownership includes Hermite normal form, the enumeration $|\mathcal{L}_N| = \sigma_1(N)$, and finite-index subgroup and subgroup-zeta theory [3]. Our bounded contribution is the temporal package for this one fixed-index shear and its recovery consequence. No priority is claimed for Hermite normal form, subgroup enumeration, subgroup-zeta theory, or Hecke correspondences. The external literature status of the exact package remains on hold.

2. Hermite coordinates and the shear action

For positive integers a, c and $0 \le b < a$, put

$$H(a,b,c) = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}, \qquad L(a,b,c) = H(a,b,c)\mathbb{Z}^2 = \mathbb{Z}(a,0) + \mathbb{Z}(b,c).$$

Proposition 2.1 (Canonical coordinates and exact action). *Every $L \in \mathcal{L}_N$ has a unique expression*

$$L = L(a,b,c), \qquad ac = N, \quad 0 \le b < a.$$

In these coordinates,

$$T_N\big(L(a,b,c)\big) = L(a, b + c \bmod a, c). \tag{1}$$

Consequently $|\mathcal{L}_N| = \sum_{a|N} a = \sigma_1(N)$.

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Proof. Let the image of L under projection onto the second coordinate be $c\mathbb{Z}$, and let $L \cap (\mathbb{Z} \times \{0\}) = a\mathbb{Z} \times \{0\}$. Choose $(b, c) \in L$ and reduce b modulo a into $0 \le b < a$. If $(x, y) \in L$, then $y = kc$ for some $k \in \mathbb{Z}$, and $(x, y) - k(b, c) \in L \cap (\mathbb{Z} \times \{0\})$. Thus $(a, 0)$ and (b, c) generate L. Projection, intersection, and the residue class of b show uniqueness. The determinant of this basis is ac, hence $ac = N$.

Left multiplication acts on the displayed column basis as

$$UH(a,b,c) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} = \begin{pmatrix} a & b+c \\ 0 & c \end{pmatrix}.$$

Reducing the first coordinate of the second generator modulo a proves (1). For each $a \mid N$ there are exactly a permitted values of b, giving the final assertion. □

Thus the divisor a labels an invariant layer, with $c = N/a$ and phase $b \in \mathbb{Z}/a\mathbb{Z}$. Define

$$g_a = \gcd(a, N/a), \qquad h_a = \frac{a}{g_a}. \tag{2}$$

3. Complete temporal census

Let $C_N(m)$ denote the number of cycles of exact length m, and let $F_N(n) = \#\operatorname{Fix}(T_N^n)$.

Theorem 3.1 (All cycles, fixed points, and zeta)**.** *For every $N, n, m \ge 1$,*

$$C_N(m) = \sum_{\substack{a \mid N \\ h_a = m}} g_a, \tag{3}$$

$$F_N(n) = \sum_{\substack{a \mid N \\ a \mid n(N/a)}} a = \sum_{\substack{a \mid N \\ h_a \mid n}} a. \tag{4}$$

In particular,

$$C_N(m) = \frac{1}{m} \sum_{d \mid m} \mu(m/d) F_N(d), \tag{5}$$

and the finite Artin–Mazur zeta function is the rational function

$$\zeta_N(z) := \exp\left(\sum_{n \ge 1} \frac{F_N(n)}{n} z^n \right) = \prod_{a \mid N} (1 - z^{h_a})^{-g_a} = \prod_{m \ge 1} (1 - z^m)^{-C_N(m)}. \tag{6}$$

The products are finite, and

$$\sum_{m \ge 1} m C_N(m) = \sigma_1(N). \tag{7}$$

Proof. On the a-layer, equation (1) is translation by $c = N/a$ on $\mathbb{Z}/a\mathbb{Z}$. Such a translation has $g_a = \gcd(a, c)$ orbits, each of length $h_a = a/g_a$. Summing these layer inventories proves (3) and (7).

The nth iterate translates by nc. It fixes a phase if and only if $a \mid nc$; when this happens, it fixes all a phases in the layer. Writing $a = g_a a'$ and $c = g_a c'$ with $\gcd(a', c') = 1$ shows that $a \mid nc$ is equivalent to $h_a = a' \mid n$, proving (4).

Every finite permutation satisfies $F_N(n) = \sum_{m \mid n} m C_N(m)$. Möbius inversion gives (5). Finally, one cycle of length m contributes

$$\exp\left(\sum_{k \ge 1} \frac{z^{km}}{k} \right) = (1 - z^m)^{-1}$$

to the zeta function. Multiplication over the cycles proves (6). □

The formulas retain the layer decomposition rather than merely the characteristic polynomial of a permutation matrix. For example, table 1 records three complete inventories.

Table 1. Exact cycle data for selected indices. An entry $m^{C_N(m)}$ means $C_N(m)$ cycles of length m.

N	$\lvert\mathcal{L}_N\rvert$	nonzero cycle inventory	$\zeta_N(z)$
8	15	$1^3,\ 2^2,\ 8^1$	$(1-z)^{-3}(1-z^2)^{-2}(1-z^8)^{-1}$
12	28	$1^3,\ 3^3,\ 4^1,\ 12^1$	$(1-z)^{-3}(1-z^3)^{-3}(1-z^4)^{-1}(1-z^{12})^{-1}$
16	31	$1^7,\ 4^2,\ 16^1$	$(1-z)^{-7}(1-z^4)^{-2}(1-z^{16})^{-1}$

4. Prime powers and the valuation staircase

The divisor layers become particularly sparse when $N = p^r$. Let $v_p(n)$ be the p-adic valuation of n.

Theorem 4.1 (Prime-power staircase)**.** *Let p be prime and $r \geq 1$. For the layer $a = p^j$, $c = p^{r-j}$, $0 \leq j \leq r$, one has*

$$g_j = p^{\min(j,r-j)}, \qquad h_j = \begin{cases} 1, & 2j \leq r, \\ p^{2j-r}, & 2j > r. \end{cases} \tag{8}$$

Hence, with

$$A_{p,r} = \sum_{j=0}^{\lfloor r/2 \rfloor} p^j = \frac{p^{\lfloor r/2 \rfloor+1} - 1}{p-1},$$

the complete cycle inventory consists of $A_{p,r}$ fixed cycles and, for each $j > r/2$, exactly p^{r-j} cycles of length p^{2j-r}. Equivalently,

$$\zeta_{p^r}(z) = (1-z)^{-A_{p,r}} \prod_{j=\lfloor r/2 \rfloor+1}^{r} (1 - z^{p^{2j-r}})^{-p^{r-j}}. \tag{9}$$

For every $n \geq 1$, put $s = v_p(n)$ and

$$J = \min\left\{ r, \left\lfloor \frac{r+s}{2} \right\rfloor \right\}.$$

Then

$$F_{p^r}(n) = \sum_{j=0}^{J} p^j = \frac{p^{J+1} - 1}{p-1}. \tag{10}$$

Thus the fixed count changes only when the valuation crosses a threshold of the same parity as r and saturates at $\sigma_1(p^r)$ when $s \geq r$.

Proof. Substitution of $a = p^j$ and $c = p^{r-j}$ into equation (2) gives $g_j = p^{\min(j,r-j)}$ and then (8). The layers with $2j \leq r$ consist entirely of fixed cycles, whose total number is $A_{p,r}$. In every remaining layer, $g_j = p^{r-j}$ and $h_j = p^{2j-r}$, proving the inventory and (9).

By (4), the jth layer contributes its p^j states precisely when $p^j \mid np^{r-j}$, or equivalently $2j \leq r + s$. Therefore the contributing layers are exactly $0 \leq j \leq J$, which gives (10). If $s \geq r$, then $J = r$. □

The parity effect is exact: both the nontrivial period exponents in base p and the fixed-count jump valuations $s = v_p(n)$ are $1, 3, \ldots, r$ for odd r and $2, 4, \ldots, r$ for even r.

5. Temporal recovery

The cycle census contains a rigid marker that is present for every index, not only at prime powers.

Theorem 5.1 (Recovery of the index)**.** *For every $N \geq 1$, the maximal cycle length of T_N is N, and there is exactly one cycle of that length. Consequently each of the following data determines N:*

(1) *the complete cycle inventory* $\{C_N(m)\}_{m\geq 1}$*;*
(2) *the complete fixed sequence* $\{F_N(n)\}_{n\geq 1}$*;*
(3) *the formal zeta series* $\zeta_N(z)$*.*

For $N = p^r$*, the recovered integer then determines both* p *and* r *by unique factorization.*

Proof. The layer $a = N$, $c = 1$ has $g_N = 1$ and $h_N = N$, so it is one N-cycle. For every divisor layer, $h_a = a/g_a \leq a \leq N$. Equality $h_a = N$ forces $a = N$ and $g_a = 1$, so the displayed cycle is the unique maximal one. Thus

$$N = \max\{m : C_N(m) > 0\}.$$

The fixed sequence determines the cycle inventory through equation (5). The formal identity

$$z\frac{d}{dz}\log\zeta_N(z) = \sum_{n\geq 1} F_N(n)z^n$$

shows coefficient by coefficient that the zeta series determines every $F_N(n)$; hence it also determines the inventory and N. The case $N = 1$ is included: there is one fixed lattice and the maximal period is 1. □

The conclusion deliberately uses temporal data. Although $|\mathcal{L}_N| = \sigma_1(N)$ is a useful accounting identity, no injectivity claim for the divisor-sum function is needed or made.

6. Exact controls and scope

The accompanying standard-library script constructs all HNF states, applies U to the two raw basis columns, reduces the off-diagonal residue, verifies mutual lattice containment, enumerates the permutation cycles, and compares the result with equation (3). For every $1 \leq N \leq 120$ it also checks every fixed count through time $2N$, Möbius reconstruction through period N, state accounting, and maximal cycle recovery. Separate symbolic lanes test theorem 4.1 for $p \in \{2, 3, 5, 7\}$ and $1 \leq r \leq 10$. All calculations use exact integers; no floating-point spectral inference enters a theorem.

The result is intentionally finite and rank two. It does not address higher-rank unipotent actions, random walks on sublattices, orbit statistics as $N \to \infty$, or the Hecke-algebraic organization of related correspondences. Those would require distinct ownership audits and proof engines.

7. Conclusion

The fixed-index sublattice action of a unipotent shear reduces exactly to divisor-indexed cyclic translations. This reduction gives a full temporal census, turns prime powers into explicit valuation staircases, and exposes the unique maximal orbit that recovers the index. The formulas are closed at finite N and are independently checkable by exhaustive integer enumeration in the accompanying artifact.

References

[1] Michael Artin and Barry Mazur. On periodic points. *Annals of Mathematics*, 81(1):82–99, 1965. doi: 10.2307/1970384.
[2] Henri Cohen. *A Course in Computational Algebraic Number Theory*, volume 138 of *Graduate Texts in Mathematics*. Springer, Berlin, Heidelberg, 1993. doi: 10.1007/978-3-662-02945-9.
[3] Fritz J. Grunewald, Daniel Segal, and G. C. Smith. Subgroups of finite index in nilpotent groups. *Inventiones Mathematicae*, 93(1):185–223, 1988. doi: 10.1007/BF01393692.